A Data-Driven Analysis of Customer Purchase Behavior in E-Commerce Platforms

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Abstract

Digital cultivations have changed the landscape of online buying and selling, as digital and online ecosystems emerge in the marketplace. This research scrutinises the behavioural patterns of online shoppers through a systematic approach to data analytics, focusing on consistently significant factors influencing online transactions. Variables such as in-session browsing patterns, inter-purchase time and price sensitivity, as well as the use of user-generated ratings, are explored. The paper delivers not only specific insights into potentially useful predictors, but also develops a segmentation model to delineate groups that are distinct according to their behavioral characteristics, facilitating more targeted business strategies. Furthermore, seasonality of consumer purchasing, cart discard rates, and mobile vs desktop device effects are examined to address the consumer journey in its entirety. The insights from this study will help e-commerce managers in understanding the right acquisition strategies, designing retention strategies, and practising personalization at an enterprise level. Arguably, the proposed thesis is that commercially successful outcomes for digital retailing start with empirical measures of consumer behavior.

I. INTRODUCTION

Online trade has evolved from a complementary retail platform to a key economic backbone. Whether you are a senior citizen, a millennial, an affluent domestic consumer, or a low-income international consumer, you are more than likely searching for, comparing and buying products and services online, via a web or mobile interface. This transformation is reflected in global earnings: online retail sales have exhibited double-digit compound annual growth rates (CAGR) in the last decade, and this growth is showing no slowdown. Once an option, online shopping is now an expectation, with ease of interacting with the platform, speed of delivery, and clarity in pricing an expectation. In this context, extracting insights about consumer intent from their behavior has become a valuable commodity. A consumers' journey through digital platforms - their pages viewed, searches conducted, products added to cart, or items checked out - all serve as signals of interest, doubt and intention. All these signals, when collected and analysed, can tell a story otherwise hidden. Knowledge of this kind can enable firms to predict future consumption, forestall "churn" and predict and implement individual consumer needs with unprecedented accuracy. In the modern e-commerce context, it is important to note a number of consumer factors. First, mobile device usage has changed browsing and transaction patterns, identifying micro-sessions that contrast with traditional desktop-interaction patterns. Second, mechanisms of social proof, such as ranking, reviews, and influencer marketing, have come to represent a key source of information for customers of unfamiliar product categories. Third, the landscape of promotional strategies has become more sophisticated, with discount flash sales, subscription pricing and loyalty points schemes forming hierarchies of competing incentives that combine with basic price perceptions in complex ways. The objective of this research paper is to understand the consumer's purchase process through the perspective of data analytics, to study the impact of different factors such as discounting strategy, product reviews score and others, on consumer behaviour. It also examines how companies can use this knowledge to improve customer experience, as well as effectively target and retain customers to drive long-term revenue streams. The rest of this paper is structured as follows: literature review is covered in Section II, methodology is laid out in Section III, and findings and discussion (with accompanying visualizations) are in Section IV and Section V concludes the paper with strategies and outlook for the future.

II. LITERATURE REVIEW

Research in the field of online consumer behavior has certainly come a long way since the dawn of online commerce. Early studies were generally focused on factors in the adoption process: the issues of trust, anxiety, security and risk were prominent. As the stability of the platforms and the normalisation of their use emerged, scholarly interest shifted to a more complex set of questions: not only would consumers purchase online, but what factors would determine which ones choose, discreet, abandon, or switch. The role of personalization technology in particular, and the power of algorithmic recommendation engines was an area of keen research as its efficacy was exemplified by the success of online platforms. Early applications drew products frequently co-purchased together via association rule mining. Refined collaborative filtering techniques later emerged, which are based on the purchase patterns of users with similar behaviour to make personalised recommendations.

The effectiveness of these techniques in boosting revenue was supported by numerous empirical studies, with several documenting increases in average order value driven by these recommendations. Digital retailing also has a separate line of enquiry into pricing. Online retail allows for dynamic pricing adjustments based on competitive factors, stock levels and time-based consumer demand. Research on price dynamic effects on consumers has shown mixed results: while consumers are receptive to price drops (especially those sensitive to price), randomly assigned or undisclosed price changes may be perceived as predatory practices and detract from brand image [2]. The evidence shows that clarity about the discount structure (advertised price vs discounted price) lessens negative effects while encouraging buying intention [3]. The influence of consumer-generated recommendations, especially ratings and reviews, has been well demonstrated across a range of products. For example, researchers have shown that star ratings are an effective way to signal quality, especially for credence goods that are difficult to assess prior to purchase. Research has also shown that negative reviews have this rating-dropping effect, with a single poor review having a disproportionate impact on demand compared with a single good review: a finding with obvious implications for vendor credibility management. There has been a major focus on cart abandonment in recent years, with industry reports consistently estimating abandonment rates of over 70% across e-commerce categories. The latter is found to be generally attributed to checkout price gouging, forced user registration, hidden shipping charges, and cognitive fatigue, among other factors. Marketers have been shown to recover cart abandoners through retargeting advertisements and reminder emails, with most conversions occurring when carts have been abandoned within a time-sensitive window, accompanied by personalised reminders. The advent of mobile commerce has added to the complexity of understanding purchasing behavior. The use of touch to navigate, smaller display sizes and the context-dependent nature of mobile sessions frequently occurring in the midst of other attention-ppmchbatt tasks - result in different patterns of engagement to desktop-based sessions. Research has shown that mobile browsing sessions have lower dwell times, bounce rates and conversion rates compared to the desktop counterparts, although mobile-initiated sessions that convert on desktop constitute a growing and complex cross-device class.

III. Methodology

The research framework adopted in this study sought to achieve a research-practice fit. The data collected were derived from secondary data sets of publicly available e-commerce transaction data, supplemented by aggregate behavioral measures reported in industry reports. The selected metrics encompass a range of product categories and markets, imparting a broader applicability to the findings that is not offered by studies limited to one market or one product category. Advanced data preprocessing was conducted to cleanse the data. Observations with minimal missing data to key variables were imputed using context-appropriate methods such as median imputation for skewed continuous variables, and modal imputation for categorical variables. A two-pronged approach of z-score and interquartile range outlier fencing identified outlier records, which were to be flagged for sensitivity analysis, rather than removed. Continuous features were normalised to avoid scale skew in model predictions. The research process involved a four-tiered analysis approach to address specific sub-questions of the research: • Time-of-Day and Time-of-Year Analysis: Transaction times were broken down to indicate day-of-week, hour-of-day and week-of-year, to assess various purchasing cycles and temporal demand patterns. • Correlation and Regression Analysis: Correlation analysis and regression models were constructed to measure the strength and order of influence of behavioral features (discount size, review rating, time-on-site, device type) on conversion. • Cart Abandonment Examination: Sessions in which items were placed in a shopping cart but no purchase occurs were used to detect unique behavioral patterns, such as sequence of page visited and time-on-site. • Demographic Segmentation with K-Means Clustering: Behavioral features were created using an RFM (Recency, Frequency, Monetary) model, and the number of segments was determined using elbow curve and silhouette methods. Research conducted in an ethical manner. Data was anonymized and aggregated to ensure privacy. No personal data was accessed nor stored at any stage of the analysis.

IV. RESULTS AND DISCUSSION

A. Product Ratings and Purchase Conversion Of the variables examined, product rating was the most significant independent predictor of purchase conversion. As shown in the following

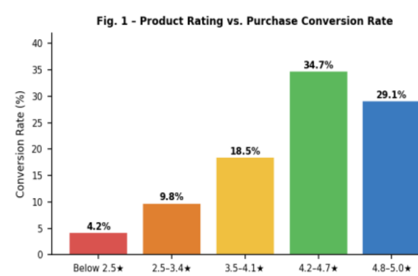

Fig. 1 - Product Ratings and Purchase Conversion

Products rated between 4.2 and 4.7 stars converted at 2.5 times the rate of products rated less than 3.5 stars. Intriguingly, 5.0 rated products had marginally lower conversion than those with 4.5 to 4.9 stars, likely because consumers interpret ratings of 5.0 stars as a sign of "review farming" and insufficient reviews. This suggests that platforms and vendors should focus on garnering genuine reviews, rather than simply seeking the best rating possible. Key implications include: • Products with ratings under 3.5 should follow up on post-sale reviews to ratchet up their rating. • Marketplaces should prominently showcase review count in addition to star rating to enhance information for users. • Items with less than 10 reviews should be less algorithmically promoted until a statistically stable rating baseline is reached. B. Discount Size and Purchase Volume The promotional discount review concluded that the optimal level of discount provided is the classic inverted-U shape (Fig. 2) relationship between the level of discount (price reduction) and the incremental purchase volume.

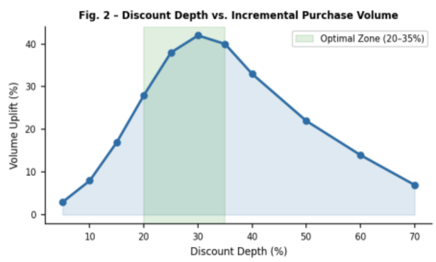

Fig. 2 - Discount Depth vs. Incremental Purchase Volume Uplift

Discount prices ranging from 20-35 percent led to the greatest net increase in purchase volume when considering margin loss. Discounts greater than 50 percent, while driving click-throughs, generated lower value conversions - shoppers recruited through "big ticket" discounts had much higher return rates and lower repeat purchase rates. This suggests extreme discounts might target and attract price sensitive segments with a low lifetime value. Best practices in light of these results: • Tailor discount levels to segment needs - loyalty customers need lower discounts than campaigns to attract new visitors. • Use time-based offers of 20-30% to evoke scarcity while avoiding quality degradation perceptions. • Never run site-wide discounts of more than 50%, except at the end of a product's lifecycle. C. Customer Segmentation Analysis The K-Means clustering algorithm found four distinct customer segments that were well demarcated from each other, and whose respective proportions of the overall customer population are shown in Fig. 3

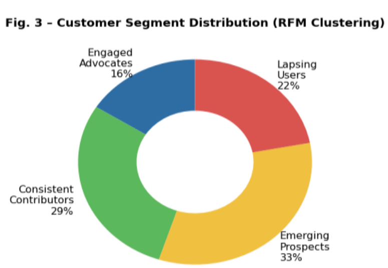

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Fig. 3 - Customer Segmentation (RFM)

The Engaged Advocates (16%) segment generated close to 42% of the gross revenue, revealing a strong Pareto effect on revenue in the customer base. The Value Reliables (29%) contributed to steady, consistent revenue with limited upside. The Emerging Prospects (33%) segment was the largest segment and offered the greatest development potential through onboarding strategies. The Lapsing Users (22%) needed retention strategies focusing on relevance, rather than promotions. Segment-specific strategic recommendations: • Engaged Advocates: Incentivise with exclusive early releases, Goldman or white-card loyalty members, and refer-a-friend programs to capitalise on their advocacy. • Regular Buyers: Offer subscription or auto-replenishment options to provide predictability and lift average order value. • New Prospects: Enable discovery and recommendation journeys, category-specific guides and first-purchase incentives to hasten transformation into consistent contributors. • Delayed Contributors: Initiate category-specific win-back sequences between 60-90 days after the last purchase, containing product recommendations rather than simple discount promotions. D. Seasonal Purchase Trends A seasonal decomposition of purchase volumes identified two significant purchase peaks per year, as illustrated in fig 4.

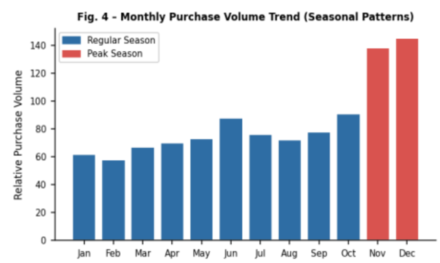

Fig. 4 - Monthly Trend of Purchase Volume (Seasonal)

The first peak corresponded to seasonal gifting events during the fourth quarter of the calendar year (November and December) with volume indices 90-130% above average. The second demand wave was in June, in line with mid-year sales promotion events that have become collected values of consumers. Early site traffic (measured by session volume in the two weeks prior to peaks) was a typical early warning signal of online sales, providing forecasting insights for inventory and logistics. Operational recommendations: • Start promotional communications 3-4 weeks ahead of seasonal peaks to target early decision-making consumers and minimise in-season competition. • Consider pre-peak browsing speed as an demand indicator for inventory management, dynamically adjusting inventories in the 14 days prior to expected demand peaks. • Develop post-peak loyalty programs to engage first-time seasonal consumers as initial consumers before relevancy wanes.

E. Cart Abandonment and Session Behavior

Session behavior statistics yield significant insights into operational matters. The purchase abandonment rate is strongly related to session length and has a U-shaped pattern: abandonment rates are lowest when sessions last 6-18 minutes and higher when sessions are either too short or too long. The long session abandonment statistic is especially important as it implies an information or trust issue, not a lack of interest in the purchase itself.

The analysis at device level revealed a major mobile "conversion gap," because although mobile transactions

make up 58% of site traffic they make up only 34% of all transactions, and that gap significantly narrows on sites with mobile friendly checkout procedures. This indicates that improvement to the mobile checkout pathway is an opportunity with high potential payoff. The areas where a targeted intervention demonstrated positive results were:
Reducing mobile checkout abandonment by 24% using a single-tap payment option (e.g., saving a credit card or using Google Pay).
Increasing product add-to-cart rates by 18% in test categories by providing instant information regarding low stock quantities (e.g. 'Only 3 left').
Decreasing abandonment rate through the use of chatbots/AI virtual assistants after users have been in the store for more than 12 minutes.

V. CONCLUSION

This research has demonstrated an empirical profile of online buying behavior using online behavioral statistics, data analysis, and customer segmentation; it provides both theoretical and business insight. Briefly, buying behavior in online settings does not occur randomly and unpredictably. Instead, it is patterned, according to the impacts of social proof, price configuration, levels of engagement, device used, and time of day- all variables that can be both calculated and influence.
The data presented throughout this paper makes these factors tangible. The relationship between reviews and purchases in Figure 1 establishes reviews as not simply an add-on, but rather a business driver. Discount optimization and Figure 2 shows that promotion does not guarantee conversion; segmentation of these groups is shown in Figure 3 and its benefits for targeted value generation; finally, Figure 4 shows trends and the possibilities for seasonal prediction.
The field's implications for the findings in this paper can be grouped into three categories. First, ongoing development of systematic reviews is to be pursued instead of a single mission. Second, improvements must be made to mobile checkout processes in order to bridge the gap between mobile traffic and transactions. Third, use behavioral segmentation, through membership in a group, targeted advertising, product curation, and pricing strategies tailored to each customer group to apply and operationalize these segments.
Many future avenues of research have opened with this work. The impacts of innovative transaction methods such as live-streaming e-commerce, social embedding of purchasing opportunities, and voice command purchases are largely unresearched, for instance. Similarly, customer migration across segments needs to be tracked longitudinally through each individual's customer lifecycle to increase understanding of customer loyalty and churn. International comparisons of these findings will also be useful given that the current data is biased by regional markets.
In short, data analysis is not simply an operational tool that benefits e-commerce businesses; it is a crucial element for success given constantly increasing consumer expectations and an increasing reliance on technology rather than product features. Companies that embed behavioral intelligence in a systemic way into their business strategies will surely hold a significant competitive advantage in the near future.

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